

Cryptanalysis of Encrypted Data to Identify Weaknesses

Tool: Hashcat / CrypTool

Theory

Cryptanalysis is the study of analysing and identifying weaknesses in cryptographic systems. Hashcat is a fast password recovery tool that tests hash strength by running dictionary and brute-force attacks against stored hashes. CrypTool is an educational platform for visualising classical and modern cipher analysis. This experiment is performed only on locally created test hashes to evaluate the strength of different hashing algorithms used in password storage.

Steps

1. Open the terminal in Kali Linux. Verify Hashcat is installed. Run: `hashcat --version` — If not installed, run: `sudo apt install hashcat -y`
 2. Generate test hashes from known passwords to use as targets. Run: `echo -n 'password' | md5sum` — Note the MD5 hash output. Run: `echo -n 'password' | sha1sum` — Note the SHA1 hash output. Run: `echo -n 'password' | sha256sum` — Note the SHA256 hash output.
 3. Save an MD5 hash to a file for Hashcat to crack. Run: `echo -n 'password' | md5sum | cut -d ' ' -f1 > hash.txt` — View the file. Run: `cat hash.txt`
 4. Decompress the rockyou wordlist if not already done. Run: `sudo gunzip /usr/share/wordlists/rockyou.txt.gz` — This wordlist contains over 14 million common passwords.
 5. Run a dictionary attack on the MD5 hash using Hashcat. Run: `hashcat -m 0 -a 0 hash.txt /usr/share/wordlists/rockyou.txt` — Here -m 0 specifies MD5 and -a 0 specifies dictionary attack mode. Hashcat finds and displays the cracked password.
 6. View the cracked result clearly. Run: `hashcat -m 0 hash.txt --show` — The output shows the hash followed by a colon and the cracked plaintext password.
 7. Test a SHA1 hash. Save it to a file. Run: `echo -n 'password' | sha1sum | cut -d ' ' -f1 > sha1hash.txt` — Run the attack with SHA1 mode. Run: `hashcat -m 100 -a 0 sha1hash.txt /usr/share/wordlists/rockyou.txt`
 8. Open CrypTool 2 from the application menu or download it from <https://www.cryptool.org> — Create a new project, type any plain text, apply a Caesar cipher with a shift of 3, and observe the encrypted output. Then use the Frequency Analysis component to crack it by identifying the most frequent letter in the ciphertext.
 9. Compare cracking times. Note how quickly MD5 and SHA1 are cracked versus attempting the same with a bcrypt hash. Run: `hashcat -m 3200 -a 0 hash.txt /usr/share/wordlists/rockyou.txt` — Observe that bcrypt is significantly slower due to its computational cost.
-

Conclusion

Hashcat successfully cracked MD5 and SHA1 hashes of weak passwords within seconds using a dictionary attack, while bcrypt proved far more resistant due to its slow hashing design. CrypTool demonstrated frequency analysis on classical ciphers. This experiment confirms that MD5 and SHA1 are unsuitable for password storage and that modern algorithms like bcrypt or Argon2 must be used instead.